

Supplementary Information:
Domain-Specific Adaptations, Mathematical Reference, and
Glossary
SCX
June 30, 2026

Contents

1	Domain-Specific Algorithm Adaptations	2
1.1	SCX Tier Selection Guide	2
2	Complete Mathematical Reference	3
2.1	Assumption Catalog (A1–A6)	3
2.2	Theorem Statements	3
2.3	Theorem Dependency Graph	4
3	Glossary of SCX Terminology	5
3.1	Notation Quick Reference	6
3.2	Typical Parameter Values	7
3.3	Cross-Reference to Paper I Supplementary Information	7

1 Domain-Specific Algorithm Adaptations

Table 1 summarizes the SCX algorithm configuration for each of the six domains surveyed in the main text. The columns correspond to the SCX protocol steps: feature representation $\phi(x)$, state discovery method, number of states K_S , number of experts M , consensus metric, threshold strategy (Tier), and validation status.

Table 1: **SCX domain-specific algorithm adaptations.** Validation status: “Validated” = experimental results available on real or benchmark data; “Proposed” = theoretical framework outlined but systematic validation pending; “Theoretical” = analysis only, no empirical data.

Domain	Subdomain	Feature $\phi(x)$	State Method	K_S	M	Thresh
Life Sciences	Drug Discovery	ECFP4/MACCS/RDKit	k-means on mol. features	5–10	4	Tier 2 (
	Medical Imaging	CNN penultimate layer	k-means on features	5–10	5–8	Tier 2 (
	Genomics (proposed)	Protein LM embeddings	k-means / spectral	4–8	4–8	Tier 2
Adv. Manufacturing	MLIP	SOAP/ACE descriptors	k-means on descriptors	4–8	8–12	Tier 3 (
	Quality control (prop.)	Sensor TCN embeddings	k-means / HMM	4–8	5–10	Tier 1/
	Process optimization	Process params + PCA	k-means	4–8	4–8	Tier 1
Embodied Intell.	Robot learning (prop.)	Trajectory autoencoders	k-means on traj. emb.	4–8	5–8	Tier 2
	Sim-to-real (prop.)	Domain-randomized feats.	k-means	4–8	5–10	Tier 2

Domain	Subdomain	Feature $\phi(x)$	State Method	K_S	M	Threshold
AI	Computer Vision	ResNet/SiCNN features	k-means on features	8–10	5–20	Tier 2/3
	NLP (proposed)	BERT/T5 embeddings	k-means	4–8	5–10	Tier 2/3
	RL (theoretical)	State-action embeddings	k-means	4–8	3–5	Tier 1
Autonomous Driv.	Perception (prop.)	LiDAR+camera features	k-means / spectral	4–8	5–10	Tier 2/3
	Behavior pred. (prop.)	Agent-history embeddings	k-means	4–8	5–8	Tier 2
	Safety (prop.)	Multi-modal fusion	k-means	4–8	10–20	Tier 3
Natural Sciences	Physics (DFT)	Functional labels + features	KB-defined	3–5	3–5	Tier 1
	Chemistry (prop.)	Molecular graphs	k-means	4–8	5–8	Tier 2
	Cryo-EM (prop.)	Micrograph patch features	k-means / spectral	4–8	3–5	Tier 1
	Earth science (prop.)	Spatial-temporal features	k-means / geo-clust.	4–8	5–20	Tier 2

1.1 SCX Tier Selection Guide

The appropriate SCX tier depends on computational budget, available domain knowledge, and noise characteristics:

[nosep]

- **Tier 0 (Loss Baseline):** One model, no SCX. Use when $\varepsilon_\phi > 0.5$, or when compute is insufficient for even $M = 5$ experts.
- **Tier 1 (SCX + Hoeffding):** $M+1$ models + k-means. Use for prototyping; simplest implementation. F1 guarantee from Theorem 1.
- **Tier 2 (SCX + Chernoff/KL):** Same compute as Tier 1; tighter bound via Chernoff information κ . Use when Tier 1’s M requirement seems too large.
- **Tier 3 (SCX + Adaptive θ_{opt}):** Same as Tier 1/2 plus η estimation. Use for best F1; critical when η is extreme (< 0.05 or > 0.5).

- **Tier 4 (SCX + Bootstrap):** Adds B bootstrap resamples. Use when η and Δ are unknown; provides diagnostic before full SCX commitment.

Minimum expert count: $M_{\min} = \max(5, \lceil \ln(1/(\eta\varepsilon_{\text{target}}))/(2\Delta^2) \rceil)$ for Tier 1, or $\max(5, \lceil \ln(1/(\eta\varepsilon_{\text{target}}))/\kappa \rceil)$ for Tiers 2–3.

2 Complete Mathematical Reference

This section summarizes the SCX mathematical theory, cross-referencing Paper I (“A Fundamental Impossibility in Data Quality”) for complete proofs. All theorems are stated without proof here; the unified theorem document [1] serves as the single source of truth.

2.1 Assumption Catalog (A1–A6)

All SCX theorems build on subsets of these six assumptions:

- A1 **Disjoint Training Sets:** Expert models are trained on disjoint data subsets $D_m \sim \mathcal{D}^{n_m}$, $D_m \cap D_{m'} = \emptyset$, $D_m \perp D_{m'}$.
- A2 **Conditional Independence (Clean):** Expert errors $\{e_m(x, y)\}_{m=1}^M$ are conditionally independent given x for clean samples.
- A3 **Bounded Loss:** The loss function $\ell(a, b) \in [0, B]$, $B < \infty$.
- A4 **Uniform Independent Noise:** Noise events are independent of x and D_m ; noise labels are uniform over $\mathcal{Y} \setminus \{y^*\}$.
- A5 **State Homogeneity:** $\sup_{x \in \mathcal{S}} \mathbb{E}[C \mid \text{clean}, X = x] \leq \mu_s$ per state s .
- A6 **Balanced Error Distribution:** $\max_{c \neq y^*} \mu_c(x) \leq C_{\text{bal}} \cdot \mu_s / (K_{\mathcal{Y}} - 1)$.

Minimal subsets that break Theorem 3’s unidentifiability: $\{A1, A4, A5\}$, $\{A1, A4, A6\}$, or $\{A5, A6\}$ with $|\mathcal{S}| \geq 2$.

2.2 Theorem Statements

Theorem 1: Noise Detection Guarantee

Under A1–A6, the SCX noise detector with state-dependent threshold θ satisfying $\mu_s < \theta < 1 - C_{\text{bal}} \cdot \mu_s / (K_{\mathcal{Y}} - 1)$ achieves:

$$\text{F1} \geq 1 - \frac{1}{\eta} \sum_{s \in \mathcal{S}} \rho_s \cdot \exp(-2M\Delta_s^2), \quad (1)$$

where $\Delta_s = \min(\theta - \mu_s, 1 - C_{\text{bal}} \cdot \mu_s / (K_{\mathcal{Y}} - 1) - \theta) > 0$ is the state-level separation gap.

Key corollary: As $M \rightarrow \infty$, $\text{F1} = 1 - O_P(\frac{1}{\eta} e^{-2M\Delta_{\min}^2})$.

Reference: Paper I, Section 1; [1], Section 1.

Theorem 2: Weak Feature Failure Lower Bound

Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^{d_\phi}$ satisfy $I(\phi(X); S) \leq \delta$ (in nats). Then:

$$\text{F1}_{\text{SCX}} \leq \text{F1}_{\text{base}} + C_F \sqrt{\delta/2}, \quad (2)$$

$$\text{AUC}_{\text{SCX}} \leq \text{AUC}_{\text{base}} + \sqrt{\delta/2} (1/\eta + 1/(1 - \eta)). \quad (3)$$

Key corollary: When $\delta / \log K_{\mathcal{S}} > 0.5$, SCX cannot outperform the loss baseline.

Reference: Paper I, Section 2; [1], Section 2.

Theorem 3: The Honest Person Theorem

For any $K \geq 2$, $M \geq 1$, and finite $\mathcal{S}$, there exist $\mathcal{P}_{\text{noise}}$ and $\mathcal{P}_{\text{hard}}$ with identical observable joint distributions such that any algorithm has minimax error $\geq \eta\rho/2$ on the ambiguous subset.

Reference: Paper I, Section 3; [1], Section 3.

Theorem 4': Exact Constant Minimax Optimality

Let $p_0 = \mu_s$, $p_1 = 1 - C_{\text{bal}} \cdot \mu_s / (K_Y - 1)$, with Chernoff information $\kappa = C(\text{Bern}(p_0), \text{Bern}(p_1))$ and Chernoff point θ^* . The SCX detector with adaptive threshold $\theta^\dagger = \theta^* + \frac{1}{M} \frac{\log((1-\eta)/\eta)}{D^*} + O(1/M^2)$ achieves:

$$\lim_{M \rightarrow \infty} e^{M\kappa} \sqrt{2\pi M} \cdot (1 - \text{F1}_{\text{SCX}}) = \frac{C_{\min}}{\eta}, \quad (4)$$

where $C_{\min} = \frac{\eta}{2} \left(\frac{1-\eta}{\eta} \right)^s \frac{1/\lambda_0^* + 1/|\lambda_1^*|}{\sqrt{\theta^*(1-\theta^*)}}$, $s = |\lambda_1^*|/D^*$, and $D^* = \log \frac{p_1(1-p_0)}{p_0(1-p_1)}$.

Matching lower bound: For any algorithm $\mathcal{A}$, $\liminf e^{M\kappa} \sqrt{2\pi M} \cdot (1 - \text{F1}_{\mathcal{A}}) \geq C_{\min}/\eta$.

Reference: Paper I, Section 4; [1], Section 4.

Theorem 5: Fixed-K State Discovery Consistency

Under sub-Gaussian noise with variance proxy σ^2 and strong separation $\Delta_{\min} = \min_{i \neq j} \|\mu_i - \mu_j\|_2 > 0$, Lloyd's k-means with $R = C_R \log n$ initializations recovers the true partition with probability:

$$P(\hat{\mathcal{C}}^{(n)} \neq \mathcal{C}^* \text{ up to permutation}) \leq K_{\mathcal{S}} \cdot \exp\left(-c_1 \frac{n_{\min} \Delta_{\min}^2}{\sigma^2 d_\phi}\right) + o(1). \quad (5)$$

Reference: Paper I, Section 5; [1], Section 5.

Proposition 6: Bootstrap Stability Diagnostic

For feature vectors $\{\phi(x_i)\}_{i=1}^N$ and $K_{\mathcal{S}}$ states, let $S(\Phi, K)$ be the mean ARI between full-data and bootstrap clusterings. Then:

- Strong features $\Rightarrow$ high stability: $S(\Phi, K) > 1 - \varepsilon$.
- No state structure $\Rightarrow$ low stability: $\mathbb{E}[S(\Phi, K)] = O(K/\sqrt{N})$.
- Diagnostic: $S(\Phi, K) < 0.7$ suggests features too weak for SCX to outperform baseline.

Reference: Paper I, Section 6; [1], Section 6.

2.3 Theorem Dependency Graph

Theorem 3 (Unidentifiability) provides necessity for $\rightarrow$ A1-A6

Theorem 1 (Noise Detection) positive result under A1-A6 $\rightarrow$ F1 $1 - O(e^{-2M^2})$

provides error model (Bernoulli(p), Bern(p))

Theorem 4' (Exact Constant) refines Thm 1's rate $\rightarrow$ exact constant

provides expert error model

Theorem 2 (Weak Feature) limits when features -weak $\rightarrow$ F1_SCX $\text{F1}_{\text{base}} + O()$

positive counterpart: strong features state discovery succeeds

Theorem 5 (Cluster Consistency) recovery of true partition $\rightarrow \exp(-c \cdot n_{\min} \cdot 2_{\min})$

practical diagnostic for Thm 2's

Proposition 6 (Stability Diag.) bootstrap stability test $\rightarrow S(K) > 0.7$

3 Glossary of SCX Terminology

Term	Definition
State (s)	A partition of the input space $\mathcal{X}$ such that samples within a state have homogeneous expert error characteristics. Discovered via k-means on feature representation $\phi(X)$. States are the fundamental unit of SCX analysis: all statistical guarantees are per-state. Denoted $s \in \mathcal{S}$, with $K_{\mathcal{S}} = \mathcal{S} $.
Consensus score ($C(x)$)	The fraction of M experts that fail to predict the given label for sample x : $C(x) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{\ell(f_m(x), y) > \tau\}$. Expected value on clean samples is μ_s (low); on noisy samples is $\approx 1 - \mu_s/(K-1)$ (high).
Separation gap (Δ_s)	For state s , the minimum distance from threshold θ to the expected consensus scores of clean and noisy samples: $\Delta_s = \min(\theta - \mu_s, 1 - C_{\text{bal}}\mu_s/(K-1) - \theta)$. Controls the exponential rate of F1 convergence.
Noise rate (η)	The global proportion of label noise in the dataset: $\eta = \mathbb{P}(\text{label flipped})$. Critical parameter for adaptive thresholding and bound validity. Typical range: 0.05–0.30.
Feature weakness (δ)	The mutual information between feature representation $\phi(X)$ and true state S : $\delta = I(\phi(X); S)$. Normalized form: $\varepsilon_{\phi} = \delta / \log K_{\mathcal{S}}$. Governs whether SCX can outperform the loss baseline.
Chernoff information (κ)	The asymptotic exponential rate of best achievable error probability for distinguishing $\text{Bern}(p_0)$ from $\text{Bern}(p_1)$: $\kappa = C(\text{Bern}(p_0), \text{Bern}(p_1)) = \min_{t \in [0,1]} \log \sum p_0^t p_1^{1-t}$. Always smaller than $2\Delta^2$.
Adaptive threshold ($\theta^\dagger$)	The noise-rate-aware threshold $\theta^\dagger = \theta^* + \frac{1}{M} \frac{\log((1-\eta)/\eta)}{D^*} + O(1/M^2)$, where θ^* is the Chernoff point and D^* is the total log-odds. Achieves minimax optimal constant.
Curation-Exploration Tradeoff	The principle that premature data cleaning (curation before exploration) destroys the signal needed to distinguish noise from hardness. Formalized as $\text{Curation}(t) = f(\text{Exploration}(t))$, with exploration rate $\eta(t)$ governing the tradeoff.
State-conditioned expertise	The property whereby different expert models develop distinct specializations across data states due to training on different subsets or with different seeds. The foundation for the SCX routing and noise detection mechanisms.
Minimax optimality (exact constant)	An algorithm is exact constant minimax optimal if it achieves the lower bound $\liminf e^{M\kappa} \sqrt{2\pi M}(1 - \text{F1}) \geq C_{\min}/\eta$ with equality—i.e., no other algorithm can achieve a smaller error constant, even asymptotically. SCX with $\theta^\dagger$ is exact constant minimax optimal.
Bootstrap ARI ($S(\Phi, K)$)	Mean adjusted Rand Index (ARI) between k-means clusterings on the full data and on bootstrap resamples. Values > 0.7 indicate strong feature structure (SCX likely beneficial); values < 0.3 indicate weak structure (SCX likely futile).
Chernoff point (θ^*)	The unique threshold that equalizes the KL divergences of clean and noisy error distributions: $\text{KL}(\theta^* \ p_0) = \text{KL}(\theta^* \ p_1)$. Also called the “minimax threshold.” Closed form: $\theta^* = \frac{\log((1-p_0)/(1-p_1))}{\log(p_1(1-p_0)/(p_0(1-p_1)))}$.

Term	Definition
Consistency score (state) ($C(s)$)	For a state s , the consistency score measures label purity within the cluster. High $C(s)$ indicates that samples within a state share the same label, suggesting the state captures a meaningful data partition.
Noise score ($\text{NS}(x)$)	Per-sample noise score combining residual and state-level metrics: $\text{NS}(x_i) = r_i/(\rho(s) + \varepsilon) \cdot (1 - C(s))$, where r_i is the residual (loss), $\rho(s)$ is the state proportion, and $C(s)$ is the state consistency.
Expert noise error rate (p_1)	The expected consensus score on noisy samples within a state. Under uniform noise (A4): $p_1 = 1 - C_{\text{bal}} \cdot \mu_s / (K_y - 1)$.
Expert clean error rate (p_0)	The expected consensus score on clean samples within a state. Equal to μ_s , the state-level clean error rate upper bound (A5).
Chernoff constant ($C_{\min}$)	The multiplicative factor in the exact minimax F1 lower bound: $C_{\min} = \frac{\eta}{2} \left(\frac{1-\eta}{\eta} \right)^s \frac{1/\lambda_0^* + 1/ \lambda_1^* }{\sqrt{\theta^*(1-\theta^*)}}$. Aggregates over multiple bottleneck states via $C_{\text{global}} = \sum_{s: \kappa_s = \kappa_{\text{global}}} \rho_s C_s$.
Total log-odds (D^*)	The sum of log-odds at the Chernoff point: $D^* = \lambda_0^* + \lambda_1^* = \log \frac{p_1(1-p_0)}{p_0(1-p_1)}$. Controls the $O(1/M)$ threshold shift.
Exponent fraction (s)	$s = \lambda_1^* /D^* \in (0, 1)$. Governs how the noise prior $((1-\eta)/\eta)^s$ factors into the minimax constant. When $\eta = 1/2$, s determines the symmetry of the threshold adaptation.

3.1 Notation Quick Reference

Symbol	Meaning	Context
$\mathcal{X}$	Input space	Domain of data samples
$\mathcal{Y}$	Label space	$ \mathcal{Y} = K_y$ classes
$\mathcal{S}$	State space	$ \mathcal{S} = K_s$ states
$s(x)$	True state assignment	Unobserved, estimated via clustering
$\phi(x)$	Feature representation	Input to state discovery
$C(x)$	Consensus score	Fraction of experts that err on x
η	Noise rate	Proportion of mislabeled samples
μ_s	State clean error bound	Expected consensus on clean samples
θ	Detection threshold	State-dependent decision boundary
Δ_s	Separation gap	Signal-to-noise ratio in state s
κ	Chernoff information	Asymptotic error exponent
M	Number of experts	Ensemble size
ρ_s	State proportion	$\mathbb{P}(X \in s)$
C_{bal}	Error balance constant	Controls error concentration (≥ 1)
ε_ϕ	Normalized feature weakness	$\delta / \log K_s$
δ	Feature-state mutual info	$I(\phi(X); S)$ in nats
θ^*	Chernoff point	Minimax threshold
$\theta^\dagger$	Adaptive threshold	η -aware optimal threshold
λ_0^*, λ_1^*	Saddlepoints	At Chernoff point
D^*	Total log-odds	$D^* = \lambda_0^* + \lambda_1^* $
s	Exponent fraction	$ \lambda_1^* /D^*$
$C_{\min}$	Minimax constant	Exact constant in F1 lower bound
$\eta(t)$	Exploration rate	Time-dependent curation aggressiveness
$S(\Phi, K)$	Bootstrap ARI	Clustering stability score
τ	Stability threshold	Heuristic = 0.7 for SCX reliability
$\Delta_{\min}$	Min cluster separation	Theorem 5's separation condition
$n_{\min}$	Min per-state samples	Theorem 5's sample condition

Symbol	Meaning	Context
σ^2	Variance proxy	Sub-Gaussian noise parameter

3.2 Typical Parameter Values

Scenario	η	Δ	Recommended M
MLIP (ACE, 500 frames)	0.10–0.20	0.25–0.50	8–12
Medical imaging (50K images)	0.05–0.10	0.30–0.50	10–15
CIFAR-10 (ResNet, 50K images)	0.08–0.10	0.35–0.50	8–10
Drug targetome (15K molecules)	0.05–0.15	0.30–0.60	4–8 (pipeline layers)
Large-scale web dataset (10M)	0.01–0.05	0.40–0.60	20–24
Small dataset ($N < 500$)	Unknown	Unknown	5–8 (bootstrap)

3.3 Cross-Reference to Paper I Supplementary Information

The full theoretical development with complete proofs is organized in Paper I’s Supplementary Information as follows:

SI Section	Content	Pages (est.)
A.1	Notation and Assumptions	2
A.2	Theorem 1: Noise Detection (Lemmas 1–3)	6
A.3	Theorem 2: Weak Feature Failure (Pinsker-TV chain)	5
A.4	Theorem 3: Unidentifiability ($K=2$ and $K>2$ constructions)	4
A.5	Theorem 4’: Exact Constant (Lemmas A–F, Bahadur-Rao, Chernoff)	12
A.6	Theorem 5: Cluster Consistency (Lemmas 1–4)	8
A.7	Proposition 6: Bootstrap Stability	3
A.8	Numerical Verification Tables	1

References

References

- [1] SCX Mathematical Theory: Unified Theorem Document. *Technical report* (2026). Available at: [\[repository\]-theory](#).
- [2] SCX Engineering Practice Guide. *Technical report* (2026).
- [3] SCX. A Fundamental Impossibility in Data Quality. *arXiv preprint* (2026).
- [4] SCX. The Curation-Exploration Tradeoff. *In preparation* (2026).
- [5] SCX. Training Data Quality Dominates Model Architecture. *In preparation* (2026).